

PROBLEM SET 8

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 30)

- 1) **(10 pts)** Recall that a particle in a one-dimensional box extending from $x = 0$ to $x = L$ is confined to the region $0 \leq x \leq L$; its wave function vanishes at the edges $x = 0$ and $x = L$ and beyond (see Exercise 5.2.5). Consider now a particle confined in a three-dimensional cubic box of volume L^3 . Choosing as the origin one of its corners, and the x , y , and z axes along the three edges meeting there, show that the normalized energy eigenfunctions are

$$\psi_E(x, y, z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_x \pi x}{L}\right) \cdot \sqrt{\frac{2}{L}} \sin\left(\frac{n_y \pi y}{L}\right) \cdot \sqrt{\frac{2}{L}} \sin\left(\frac{n_z \pi z}{L}\right)$$

with energy $E = \frac{\hbar^2 \pi^2}{2ML^2} (n_x^2 + n_y^2 + n_z^2)$, and n_i are positive integers.

10.2.1.

Inside the box, $V = 0$. The Schrödinger Equation:

$$E \psi(x, y, z) = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi(x, y, z)$$

Separation of variables:

$$\psi(x, y, z) = X(x)Y(y)Z(z)$$

$$\Rightarrow E X(x)Y(y)Z(z) = -\frac{\hbar^2}{2m} \cdot Y(y)Z(z) \frac{\partial^2}{\partial x^2} X(x) - \frac{\hbar^2}{2m} X(x)Z(z) \frac{\partial^2}{\partial y^2} Y(y) - \frac{\hbar^2}{2m} X(x)Y(y) \frac{\partial^2}{\partial z^2} Z(z)$$

Divide both sides by $X(x)Y(y)Z(z)$

$$\Rightarrow E = -\frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2}{\partial x^2} X - \frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2}{\partial y^2} Y - \frac{\hbar^2}{2m} \frac{1}{Z} \frac{\partial^2}{\partial z^2} Z$$

The only way to satisfy this equation is to set

$$\begin{cases} -\frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2}{\partial x^2} X = E_x \\ -\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2}{\partial y^2} Y = E_y \\ -\frac{\hbar^2}{2m} \frac{1}{Z} \frac{\partial^2}{\partial z^2} Z = E_z \end{cases} \quad E = E_x + E_y + E_z$$

We have got 3 one-dimensional Schrödinger Equation. The solutions are:

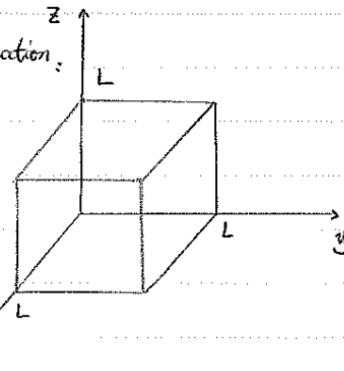
$$\begin{cases} X(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_x \pi x}{L}\right), & E_x = \frac{\hbar^2 \pi^2}{2ML^2} n_x^2 \\ Y(y) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_y \pi y}{L}\right), & E_y = \frac{\hbar^2 \pi^2}{2ML^2} n_y^2 \\ Z(z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_z \pi z}{L}\right), & E_z = \frac{\hbar^2 \pi^2}{2ML^2} n_z^2 \end{cases}$$

$$n_x, n_y, n_z = 1, 2, 3, \dots$$

Therefore,

$$\psi_E(x, y, z) = X_E(x)Y_E(y)Z_E(z) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_x \pi x}{L}\right) \sqrt{\frac{2}{L}} \sin\left(\frac{n_y \pi y}{L}\right) \sqrt{\frac{2}{L}} \sin\left(\frac{n_z \pi z}{L}\right),$$

$$E = E_x + E_y + E_z = \frac{\hbar^2 \pi^2}{2ML^2} (n_x^2 + n_y^2 + n_z^2).$$



2) (10 pts) Quantize the two-dimensional oscillator for which

$$\mathcal{H} = \frac{p_x^2 + p_y^2}{2m} + \frac{1}{2}m\omega_x^2 x^2 + \frac{1}{2}m\omega_y^2 y^2$$

hint: quantize, i.e., replacing the classical variables with quantum operators; you may refer to general results from section 7.3 (Eq. 7.3.22)

(a) (5 pts) Show that the allowed energies are

$$E = (n_x + 1/2)\hbar\omega_x + (n_y + 1/2)\hbar\omega_y, \quad n_x, n_y = 0, 1, 2, \dots$$

10.2.2 (1) Two dimensional oscillator.

$$\begin{aligned} H &= \left(\frac{p_x^2}{2m} + \frac{1}{2}m\omega_x^2 x^2 \right) + \left(\frac{p_y^2}{2m} + \frac{1}{2}m\omega_y^2 y^2 \right) \\ &= H_x + H_y \end{aligned}$$

Similar to the last problem, we can separate variables

$$\psi_{n_x n_y}(x, y) = \psi_{n_x}(x) \psi_{n_y}(y)$$

$$\Rightarrow (H_x + H_y) \psi_{n_x}(x) \psi_{n_y}(y) = (E_x + E_y) \psi_{n_x}(x) \psi_{n_y}(y).$$

$$\Rightarrow H_i \psi_{n_i} = E_{n_i} \psi_{n_i} \Rightarrow \begin{cases} \psi_{n_i} = A_{n_i} H_{n_i} \left(\sqrt{\frac{m\omega}{\hbar}} x_i \right) e^{-\frac{m\omega x_i^2}{2\hbar}} \\ E_{n_i} = \hbar\omega_i \left(n_i + \frac{1}{2} \right), n_i = 0, 1, 2, \dots \end{cases}$$

$$\Rightarrow E = E_{n_x} + E_{n_y} = \hbar\omega_x \left(n_x + \frac{1}{2} \right) + \hbar\omega_y \left(n_y + \frac{1}{2} \right)$$

- (b) **(5 pts)** Write down the corresponding wave functions in terms of single oscillator wave functions. Verify that they have definite parity (even/odd) number $x \rightarrow -x, y \rightarrow -y$ and that the parity depends only on $n = n_x + n_y$.

$$(2) \quad \psi_{n_x n_y} = A_{n_x} A_{n_y} H_{n_x} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) H_{n_y} \left(\sqrt{\frac{m\omega}{\hbar}} y \right) e^{-\frac{m\omega}{2\hbar} (x^2 + y^2)}$$

where $A_{n_i}^2 = \sqrt{\frac{m\omega}{\pi\hbar 2^{2n_i} (n_i)!^2}}$ is the normalization factor

$$\Rightarrow \psi_{n_x n_y}(-x, -y) = \left[A_{n_x} A_{n_y} e^{-\frac{m\omega}{2\hbar} (x^2 + y^2)} \right] H_{n_x} \left[-\sqrt{\frac{m\omega}{\hbar}} x \right] H_{n_y} \left[-\sqrt{\frac{m\omega}{\hbar}} y \right]$$

Since $H_{n_i} \left(-\sqrt{\frac{m\omega}{\hbar}} x_i \right) = (-1)^{n_i} H_{n_i} \left[\sqrt{\frac{m\omega}{\hbar}} x_i \right],$

$$\psi_{n_x n_y}(-x, -y) = (-1)^{n_x + n_y} \psi_E(x, y), \text{ so the parity depends on } n \equiv n_x + n_y.$$

- 3) (5 pts) Two identical bosons are found to be in states $|\phi\rangle$ and $|\psi\rangle$. Show that the normalized state vector describing the system when $\langle\phi|\psi\rangle \neq 0$ is

$$|\lambda\rangle = \frac{|\phi\psi\rangle + |\psi\phi\rangle}{\sqrt{2(1 + |\langle\phi|\psi\rangle|^2)}}$$

10.3.1 When these two bosons form a system,

$$|\lambda\rangle = A (|\psi\phi\rangle + |\phi\psi\rangle).$$

Here $|\psi|^2 = 1$, $|\phi|^2 = 1$, but $\langle\psi|\phi\rangle \neq 0$, so.

$$\begin{aligned} 1 &= \langle\lambda|\lambda\rangle = |A|^2 [\langle\psi\phi|\psi\phi\rangle + \langle\psi\phi|\phi\psi\rangle + \langle\phi\psi|\psi\phi\rangle + \langle\phi\psi|\phi\psi\rangle] \\ &= |A|^2 [\langle\psi|\psi\rangle\langle\phi|\phi\rangle + \langle\psi|\phi\rangle\langle\phi|\psi\rangle + \langle\phi|\psi\rangle\langle\psi|\phi\rangle + \langle\phi|\phi\rangle\langle\psi|\psi\rangle] \\ &= |A|^2 [1 + 2|\langle\phi|\psi\rangle|^2 + 1] \end{aligned}$$

$$\Rightarrow |A| = \frac{1}{\sqrt{2(1 + |\langle\phi|\psi\rangle|^2)}}, \text{ we can choose the phase } 0$$

$$\Rightarrow |\lambda\rangle = \frac{|\psi\phi\rangle + |\phi\psi\rangle}{\sqrt{2(1 + |\langle\phi|\psi\rangle|^2)}}$$

- 4) (5 pts) When an energy measurement is made on a system of three bosons in a box, the n values obtained were 3, 3, and 4. Write down a symmetrized, normalized state vector $|\psi\rangle$.

10.3.2. Here we have $|3\rangle, |3\rangle, |4\rangle$, where

$$\langle 3|3\rangle = \langle 4|4\rangle = 1, \quad \langle 3|4\rangle = 0.$$

$$\Rightarrow |\lambda\rangle = \frac{1}{\sqrt{3}} (|3,3,4\rangle + |3,4,3\rangle + |4,3,3\rangle)$$

$\langle \lambda|\lambda\rangle = 1$, and $|\lambda\rangle$ is invariant under permutations of any two particles.